Lecture 1

Spaces and continuity

Objectives. *Continuity can be described without measuring distances. We keep the open sets of a metric space and use them to define a topological space. We then prove that metric continuity is equivalent to openness of inverse images. This gives a way to compare spaces and to put a topology on a subset or a product. Explicit maps identify an interval with a line and an open disc with a plane.*

By the end of this lecture you will be able to:

- Construct a topology from a metric or a basis.
- Test continuity by inverse images and verify a homeomorphism.
- Use the continuity tests for subspaces and finite products.
- Separate points in a Hausdorff space and prove uniqueness of sequence limits.

1.1 Open sets from a metric

Recall the distance function from *Linear Algebra for Physics*, Lecture 13. A *metric* on a set X is a function $d\colon X \times X \to [0,\infty)$ such that $d(x,y) = 0$ exactly when $x = y$, $d(x,y) = d(y,x)$, and $d(x,z) \le d(x,y) + d(y,z)$ for all $x, y, z \in X$. The last condition is the triangle inequality. The *open ball* with centre $x \in X$ and radius $r > 0$ is

$$B_r(x) = \{y \in X : d(x,y) < r\}.$$

A subset $U \subseteq X$ is *open* if every $x \in U$ has a ball $B_r(x) \subseteq U$ with $r > 0$. The radius may depend on the point.

For example, in $\mathbb{R}$ with $d(x,y) = |x - y|$, an interval (a,b) is open. At $x \in (a,b)$ take $r = \min(x - a, b - x) > 0$. Neither $[a,b]$ nor $[a,b)$ is open when $a < b$, because no ball about a stays in the interval. The empty set is open because it has no points to test.

Every point of an open ball has a smaller ball about it inside the original ball. The triangle inequality explains the inclusion in the figure.

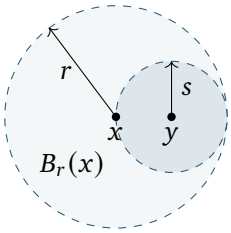


Figure 1.1: If $y \in B_r(x)$, the ball about y of radius $s = r - d(x,y)$ lies inside $B_r(x)$. The smaller boundary is also excluded.

Indeed, if $z \in B_s(y)$, then

$$d(x,z) \le d(x,y) + d(y,z) < d(x,y) + s = r.$$

Thus every open ball is open.

Proposition 1.1 (Open sets of a metric space)**.** *The open subsets of a metric space include $\varnothing$ and X. Arbitrary unions and finite intersections of open sets are open.*

Proof. The whole set contains every ball. If x belongs to a union of open sets, it belongs to one member of the union. That member contains a ball about x, so the union does too. If $x \in U \cap V$ with U and V open, choose $B_r(x) \subseteq U$ and $B_s(x) \subseteq V$. Then $B_{\min(r,s)}(x) \subseteq U \cap V$. Repeating this argument gives any nonempty finite intersection. The intersection of no subsets is X. ∎

The restriction to finite intersections matters. In $\mathbb{R}$ the intersection

$$\bigcap_{n \ge 1} (-1/n, 1/n) = \{0\}$$

is not open.

1.2 Topological spaces and bases

Definition 1.2 (Topology)**.** A *topology* on a set X is a collection $\mathcal{T}$ of subsets of X that contains $\varnothing$ and X and is closed under arbitrary unions and finite intersections. The pair $(X, \mathcal{T})$ is a *topological space*. Its members are called *open sets*. A subset is *closed* if its complement in X is open.

The preceding proposition constructs the *metric topology* of d. The standard topology on $\mathbb{R}^n$ is the topology of the Euclidean distance. We use this topology unless another one is specified. The same set may carry other topologies. The *discrete topology* contains every subset. The *indiscrete topology* contains only $\varnothing$ and X. Both satisfy the axioms directly.

A set can be both open and closed. The sets X and $\varnothing$ always are. A set can be neither open nor closed. The set $[0,1)$ in the standard real line is one example. Its complement is not open at 1.

For a subset $A \subseteq X$, its *interior* $\operatorname{int} A$ is the union of all open subsets of A. Its *closure* $\overline{A}$ is the intersection of all closed sets containing A. The interior is open by the union axiom. The closure is closed because its complement is a union of open sets. Every open subset of A is one of the sets being united, so lies in $\operatorname{int} A$. Every closed superset of A contains the defining intersection, so contains $\overline{A}$.

A *neighbourhood* of $x \in X$ is a set containing an open set that contains x. A neighbourhood itself need not be open.

In the standard real line, $\operatorname{int}[0,1) = (0,1)$ and $\overline{[0,1)} = [0,1]$. To check the second equality, note that every open interval about 1 meets $[0,1)$, so no closed superset can omit 1. The closed set $[0,1]$ itself contains $[0,1)$.

Listing all open sets is usually unnecessary. Open balls already determine the metric topology because every open set is a union of balls.

Definition 1.3 (Basis)**.** A collection $\mathcal{B}$ of subsets of X is a *basis* if it covers X and has the following refinement property: whenever $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, there is $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Proposition 1.4 (Topology from a basis). *The unions of members of a basis form a topology. A set U belongs to this topology exactly when every $x \in U$ lies in some $B \in \mathcal{B}$ with $B \subseteq U$.*

Proof. The empty union is $\varnothing$, and the covering property gives X. A union of unions is a union. For two unions U and V, a point $x \in U \cap V$ lies in basis members $B_1 \subseteq U$ and $B_2 \subseteq V$. Refine their intersection to B_3 as in the definition. The union of all these refinements is $U \cap V$. Finite intersections follow by repetition. The final assertion is precisely the condition that U be a union of basis members. ∎

The collection $\mathcal{B}$ is a *basis for a topology* $\mathcal{T}$ when $\mathcal{T}$ is exactly the topology it generates. Open balls form a basis for the metric topology. They cover the space, and the open intersection of two balls contains a ball about each of its points.

For example, open intervals form a basis for $\mathbb{R}$. Intervals with rational endpoints suffice: for $x \in (a, b)$ choose rational p and q with $a < p < x < q < b$.

The rational intervals are countable. A space is *second countable* if its topology has a countable basis.

In $\mathbb{R}^n$, boxes with rational endpoints give a countable basis as well. Every open ball contains such a box about each of its points, and every open box contains a ball about each of its points. Thus boxes and balls give the same topology. Second countability will be a condition in the definition of a manifold in Lecture 7.

1.3 Continuity and homeomorphisms

Definition 1.5 (Continuity). A map $f : X \to Y$ of topological spaces is *continuous* if $f^{-1}(V)$ is open in X for every open set $V \subseteq Y$. Here the *inverse image* is the set

$$f^{-1}(V) = \{x \in X : f(x) \in V\}.$$

An inverse image exists for every map. The notation does not assume that the map is invertible. It is enough to test basis members of Y, since inverse images preserve unions.

The inverse image can have several pieces even when the target set is an interval. For the map $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, we obtain

$$f^{-1}((1, 4)) = (-2, -1) \cup (1, 2).$$

The figure shows the two pieces. Both are open, as required by continuity.

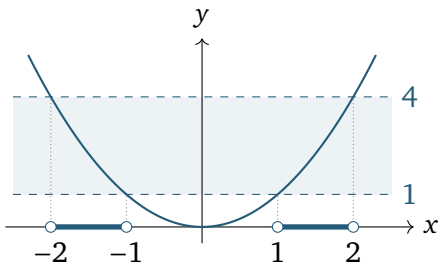


Figure 1.2: The horizontal band $1 < y < 4$ meets the graph $y = x^2$ above two open intervals on the x-axis. Endpoints are excluded.

Proposition 1.6 (Metric and topological continuity). *For metric spaces X and Y, the map $f : X \to Y$ is continuous exactly when, for every $x \in X$ and every $\varepsilon > 0$, there is $\delta > 0$ such that*

$$d_X(x, x') < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x')) < \varepsilon \qquad (x' \in X).$$

Proof. Suppose inverse images of open sets are open. The inverse image of $B_\varepsilon(f(x))$ is an open set containing x. It contains a ball $B_\delta(x)$, which gives the implication.

Conversely, suppose the displayed condition holds. For an open set $V \subseteq Y$ and $x \in f^{-1}(V)$ choose $B_\varepsilon(f(x)) \subseteq V$. The condition supplies a ball $B_\delta(x) \subseteq f^{-1}(V)$. Thus $f^{-1}(V)$ is open. ∎

Continuity therefore depends on a metric only through its open sets. For example, two norms satisfying $c\,\|v\|_a \le \|v\|_b \le C\,\|v\|_a$ for every $v \in \mathbb{R}^n$, with $c, C > 0$, induce the same topology. Indeed, their balls satisfy

$$B^a_{r/C}(x) \subseteq B^b_r(x) \subseteq B^a_{r/c}(x).$$

Applying these inclusions at each point gives the same open sets. These are called *equivalent norms*. Equivalent norms give the same continuous maps, even though they may assign different lengths. Lecture 7 uses this when comparing charts.

Identity maps are continuous, and composites of continuous maps are continuous. The latter follows from $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$ for maps $f\colon X \to Y$ and $g\colon Y \to Z$ and an open set $W \subseteq Z$.

Definition 1.7 (Homeomorphism)**.** A *homeomorphism* is a continuous bijection $f\colon X \to Y$ whose inverse $f^{-1}\colon Y \to X$ is continuous. Spaces joined by such a map are *homeomorphic*, written $X \cong Y$.

For example, $f\colon (0,1) \to \mathbb{R}$ given by $f(t) = \tan(\pi(t - \frac{1}{2}))$ is a homeomorphism. Its inverse is given by $f^{-1}(s) = \frac{1}{2} + \pi^{-1} \arctan s$. Both maps are continuous by ordinary one-variable calculus and the metric-continuity proposition.

1.4 Subspaces and products

The circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a subset of the plane. Its open sets must describe motion along the circle, although no plane ball lies inside the circle.

Definition 1.8 (Subspace and product topologies)**.** The *subspace topology* on $A \subseteq X$ consists of the sets $A \cap U$ with U open in X. For topological spaces X and Y, the *product topology* on $X \times Y$ has basis the rectangles $U \times V$ with $U \subseteq X$ and $V \subseteq Y$ open. The topology on a finite product has basis the products of open sets in its factors.

Intersections with A preserve unions and finite intersections, so the subspace sets form a topology. Rectangles cover the product, and the intersection of two rectangles is a rectangle. Hence the basis proposition constructs the product topology.

The restricted metric has balls $A \cap B_r(x)$ for $x \in A$. They are subspace-open. Conversely, if $x \in A \cap U$ with U open in X, a ball $B_r(x) \subseteq U$ gives a restricted ball inside $A \cap U$. Thus the two topologies agree.

For $\mathbb{R}^n$, every product of open sets is a union of open boxes, by choosing an interval about each coordinate. The box-and-ball comparison therefore identifies the finite product and Euclidean topologies.

The figure compares the two constructions. An open arc on the circle is open relative to the circle. An open rectangle in the plane is open in the product of two real lines.

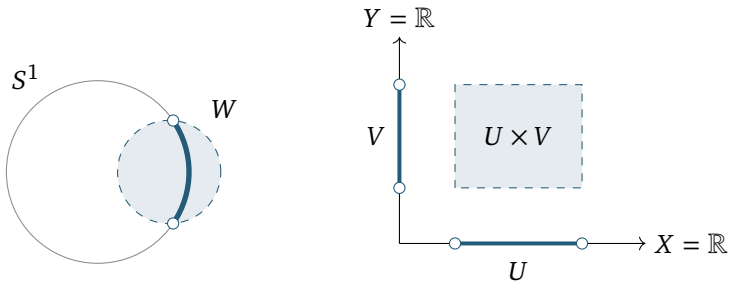


Figure 1.3: An open disc W cuts out an open arc on the circle. Open intervals U and V determine an open rectangle $U \times V$ in the plane.

Proposition 1.9 (Continuity tests for subspaces and products)**.** *For a map $f\colon Z \to A \subseteq X$, continuity into the subspace A is equivalent to continuity of $i \circ f\colon Z \to X$, where $i\colon A \to X$ is inclusion. A map $h\colon Z \to X \times Y$ is continuous exactly when its components $\pi_X \circ h$ and $\pi_Y \circ h$ are continuous. Here $\pi_X(x, y) = x$ and $\pi_Y(x, y) = y$.*

Proof. For an open set $U \subseteq X$, the inverse image under inclusion is $i^{-1}(U) = A \cap U$, so i is continuous. Continuity of f then gives continuity of the composite. Conversely, $f^{-1}(A \cap U) = (i \circ f)^{-1}(U)$, proving continuity into A.

The projections are continuous because $\pi_X^{-1}(U) = U \times Y$ and $\pi_Y^{-1}(V) = X \times V$ are open. Thus continuous h has continuous components. Conversely, for a basis rectangle the inverse image is

$$h^{-1}(U \times V) = (\pi_X \circ h)^{-1}(U) \cap (\pi_Y \circ h)^{-1}(V).$$

Continuous components make this intersection open. Testing basis members proves continuity of h. ∎

In particular, restricting a continuous map $f\colon X \to Y$ to $A \subseteq X$ preserves continuity, since the restriction is $f \circ i$.

For example, the map $t \mapsto (\cos t, \sin t)$ is continuous into $\mathbb{R}^2$ because both components are continuous. Its values lie in S^1, so the subspace test makes it continuous into the circle too. The same proof, with one projection and one inverse image per factor, gives the test for every finite product.

A continuous bijection need not have continuous inverse. Restrict the preceding map to $f\colon [0, 2\pi) \to S^1$. Each point of the circle has exactly one angle in this interval, so f is bijective. The set $[0, \pi)$ is open in $[0, 2\pi)$, but its image is not open in the circle. Every open neighbourhood of $(1, 0)$ in the circle contains points just below the horizontal axis. Thus the inverse map is not continuous. Lecture 2 supplies a useful sufficient condition for a continuous bijection to be a homeomorphism.

1.5 Hausdorff spaces and invariants

Definition 1.10 (Hausdorff space)**.** A space X is *Hausdorff* if any two distinct points have disjoint open neighbourhoods. A sequence $(x_n)_{n\geq 1}$ *converges* to $x \in X$ if every open neighbourhood of x contains all x_n for sufficiently large n.

Proposition 1.11 (Separation and limits)**.** *Metric spaces are Hausdorff. Subspaces and finite products of Hausdorff spaces are Hausdorff. A sequence in a Hausdorff space has at most one limit.*

Proof. For distinct points x and y of a metric space, put $r = d(x,y)/3$. The balls $B_r(x)$ and $B_r(y)$ are disjoint. A point in both would give $d(x,y) < 2r < d(x,y)$ by the triangle inequality. Intersecting disjoint open neighbourhoods with a subspace separates its points. Two distinct points in a product differ in some coordinate. Separate those coordinate values and take inverse images under the corresponding projection. The resulting product neighbourhoods are disjoint.

If a sequence had distinct limits x and y, disjoint open neighbourhoods U and V of those points would both contain every term after some common index. This would put a term in $U \cap V = \varnothing$. ∎

The Hausdorff condition can fail. The *cofinite topology* on $\mathbb{R}$ consists of $\varnothing$ and the subsets with finite complement. It is a topology. A nonempty union has complement contained in the finite complement of any one nonempty member. A finite intersection of nonempty open sets has complement a finite union of finite sets. Two nonempty open sets cannot be disjoint, since their intersection has finite complement in the infinite set $\mathbb{R}$. This space is not Hausdorff.

A *topological invariant* is a property shared by homeomorphic spaces. Hausdorffness is one example.

Proposition 1.12 (Transport of open sets)**.** *A homeomorphism $f : X \to Y$ gives a bijection $U \mapsto f(U)$ between their topologies, preserving unions, intersections and inclusion. Hausdorffness and second countability are topological invariants.*

Proof. If $f : X \to Y$ is a homeomorphism and $U \subseteq X$ is open, then $f(U) = (f^{-1})^{-1}(U)$ is open by continuity of the inverse. Continuity of f gives the converse correspondence. A bijection preserves the stated set operations. Images of disjoint open neighbourhoods separate the image points, proving Hausdorff invariance. If $\mathcal{B}$ is a countable basis for X, its images form a countable family of open sets. Bijectivity carries the covering and refinement properties across. Every open set of Y is the image of a union of members of $\mathcal{B}$, hence a union of these image sets. They form a basis for Y. ∎

The standard real line and the cofinite real line are therefore not homeomorphic. They have the same underlying set but different separation properties.

1.6 Explicit homeomorphisms

We compare the open interval, the line, a circle with a point removed, the open disc and the plane. All subsets have their subspace topologies. The interval and the line were identified above. To treat the punctured circle, remove $N = (0,1)$ and define

$$p : S^1 \setminus \{N\} \longrightarrow \mathbb{R}, \qquad p(x,y) = \frac{x}{1-y}.$$

The line through N and (x,y) meets the horizontal axis at $(p(x,y),0)$. This construction is called *stereographic projection*. The figure shows this construction.

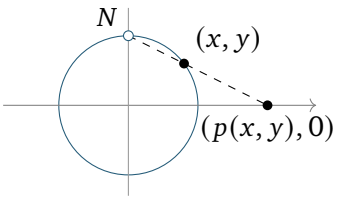


Figure 1.4: Stereographic projection from N onto the horizontal axis. The projection point N is omitted from the circle.

Solving the line and circle equations gives the inverse

$$p^{-1}\colon \mathbb{R} \longrightarrow S^1 \setminus \{N\}, \qquad p^{-1}(t) = \left(\frac{2t}{1+t^2}, \frac{t^2-1}{1+t^2}\right).$$

The squared coordinates sum to 1, and the second coordinate is less than 1. Substitution gives $p(p^{-1}(t)) = t$. Conversely, using $x^2 = 1 - y^2$ and $y \neq 1$ gives $p^{-1}(p(x,y)) = (x,y)$. Both maps have continuous coordinate functions with nonzero denominators on their domains. The subspace and product tests therefore make p a homeomorphism. By composition the punctured circle is homeomorphic to $(0,1)$ as well.

For the open disc $D = \{u \in \mathbb{R}^2 : \|u\| < 1\}$, move each point outward along its radius. Define

$$f_D\colon D \to \mathbb{R}^2, \qquad f_D(u) = \frac{u}{1-\|u\|}, \qquad f_D^{-1}\colon \mathbb{R}^2 \to D, \qquad f_D^{-1}(v) = \frac{v}{1+\|v\|}.$$

The norm here is Euclidean and continuous. The denominators are positive, so both formulas are continuous, including at 0. For $r = \|u\| < 1$, the norm of $f_D(u)$ is $r/(1-r)$, giving $f_D^{-1}(f_D(u)) = u$. For $s = \|v\|$, the inverse formula has norm $s/(1+s) < 1$, so lands in D. It gives $f_D(f_D^{-1}(v)) = v$. The subspace test proves continuity into D. Thus f_D is a homeomorphism.

The results proved so far form two families:

Spaces identified	Maps used
$(0,1) \cong \mathbb{R} \cong S^1 \setminus \{N\}$	tangent and stereographic projection
$D \cong \mathbb{R}^2$	radial map f_D

Nothing proved here distinguishes the two families. Lecture 2 proves that a line and a plane are not homeomorphic by examining what happens when a point is removed. It also distinguishes the circle from the line by compactness, and the circle from the closed interval by removing a point. Those distinctions are not assumed here.

Self-check. Here x^i and u^i label coordinates. A squared coordinate is written $(x^i)^2$. For the unit sphere $S^2 = \{(x^1,x^2,x^3) \in \mathbb{R}^3 : (x^1)^2 + (x^2)^2 + (x^3)^2 = 1\}$ and $N = (0,0,1)$, prove that the map

$$p_N\colon S^2 \setminus \{N\} \to \mathbb{R}^2, \qquad p_N(x) = \frac{(x^1, x^2)}{1 - x^3}$$

is a homeomorphism by finding its inverse.

Answer. For $u = (u^1, u^2) \in \mathbb{R}^2$, put $r^2 = (u^1)^2 + (u^2)^2$. The inverse is

$$p_N^{-1}(u) = \frac{(2u^1, 2u^2, r^2 - 1)}{1 + r^2}.$$

Its squared norm is $[4r^2 + (r^2 - 1)^2]/(1 + r^2)^2 = 1$ and its last coordinate is less than 1. Substitution gives $p_N(p_N^{-1}(u)) = u$. For $x \in S^2 \setminus \{N\}$, the sphere equation gives $\|p_N(x)\|^2 = (1 + x^3)/(1 - x^3)$, hence $p_N^{-1}(p_N(x)) = x$. The coordinate formulas and the subspace and product tests prove continuity of both maps.

Summary. A topology records which subsets are open. Continuity means that inverse images of open sets are open, and agrees with metric continuity when a metric is present. A homeomorphism requires continuity in both directions. Subspace and product topologies let us test continuity using inclusion and coordinate projections. Hausdorffness is preserved by homeomorphisms and ensures uniqueness of sequence limits. Lecture 2 introduces compactness and connectedness to distinguish more spaces.

Reading. Munkres, *Topology*, second edition, §§12–13 and 15–18: topologies, bases, products, subspaces, separation and continuity.